Digital Mohalla — Class diagram, Student 3 packages (design step A7, v1.0): Module 9 cases, decisions and audit; Module 10 scenario suite

Python design classes (iteration 1), from UC-D6, UC-I4, UC-I6, the escalation-case state machine and schema.sql. Each class lists the FRs it realises. Other students' classes appear only as boundary interfaces (blue S1, orange S2, grey shared).

moderation (Module 9)

«interface» IModeration

queue(community_id, filters): list[CaseSummary]
case_detail(case_id): CaseDetail
request_information(case_id, request, deadline)
record_decision(case_id, kind, reason, ...)
audit_log(filters): list[AuditEntry]

FR-T-10, FR-T-11, FR-M-05 (REST: /cases, /audit-log)

«interface» ICaseIntake

open_or_attach(transaction_id, trigger, reason): EscalationCase
withdraw(transaction_id)

FR-T-10, UC-D6 1b (one live case per transaction)

AuditLogService

record(actor, action, target, outcome, reason?)
query(filters): list[AuditEntry] {the view is logged}
verify_chain(): bool
mask(entry): AuditEntry

FR-M-05, NFR-S-05, UC-I6

«value» AuditEntry

occurred_at, actor_id, authorisation
action, target_type, target_id
previous_value, new_value (masked)
outcome: SUCCESS | REJECTED, reason
prev_hash, row_hash

FR-M-05 (audit_log row)

«use»

CaseService

- cases: CaseRepository
- notifier: INotify

open_or_attach(transaction_id, trigger, reason)
queue(community_id, filters)
case_detail(case_id)
request_information(case_id, request, deadline)
add_statement(case_id, party, text, attachments)
withdraw(transaction_id)
route_to_admin(case_id)

FR-T-10, FR-T-11 (step 2), FR-B-17, UC-D6 2a

«use»

EvidenceTrailBuilder

- events: IEventQuery

build(case): EvidenceTrail
gaps(trail): list[str]

FR-T-10, UC-D6 1, 1a (trail derived from the event log)

DecisionService

- lifecycle: ILifecycleCommands
- audit: AuditLogService

record_decision(case_id, moderator, kind, reason, safeguards?, resolved_outcome?) : ModeratorDecision
allowed_kinds(case): set[DecisionKind] {ADR-0003}

FR-T-11, ADR-0003, ADR-0004, UC-I4 3c, 4a

«enumeration» CaseTrigger, CaseStatus, DecisionKind

CaseTrigger: DISPUTE | MANIPULATION_FLAG | HIGH_VALUE_LOW_CONFIDENCE | LOCALITY_HOLD | ENGINE_FAILURE
CaseStatus: OPEN | AWAITING_INFORMATION | DECIDED | APPEALED | REVERSED | WITHDRAWN
DecisionKind: CLEARED | SAFEGUARDS_IMPOSED | RESTRICTED | DISPUTE_RESOLVED
(appeal, reversal kinds: iteration 2)

1 manages 0..*

«creates»

«creates»

EscalationCase

id, transaction_id: UUID
opened_event_id: int
trigger_type: CaseTrigger
hold_reason | dispute_reason
raised_by: UUID | None
status: CaseStatus
priority: int {1..3}
reason: str
assigned_to: UUID | None
info_deadline, opened_at, closed_at

assign(moderator)
await_information(deadline)
mark_no_response()
close(decision | withdrawn)

FR-T-10; state machine C01–C11

«value» EvidenceTrail

events: list[LifecycleEvent]
assessment: AssessmentResult | None
condition_reports, statements
attachments (metadata stripped)
gaps: list[str]

FR-T-10, NFR-PR-03 (chat only if disputed)

ModeratorDecision

id, case_id: UUID
decided_by: UUID, decider_role
kind: DecisionKind
reason: str {mandatory}
safeguards: list
resolved_outcome: dict | None
reverses_decision_id: UUID | None
decided_at: datetime

FR-T-11, FR-B-15 (resolved outcome is evidence), ES-11

1 — decided by — 0..*

1 — 0..*

CaseStatement

author_id, author_role
statement: str
created_at

FR-B-17, UC-B12 5, UC-I4 2

Notes

- Module 9 changes a lifecycle state only through ILifecycleCommands.apply_decision() (one writer of lifecycle events, A8); the allowed decision kinds depend on the case type and hold reason (ADR-0003).
- The evidence trail is derived from the event log, never stored separately (UC-D6).
- ICaseIntake is realised inside the same database transaction as the hold or dispute event in iteration 1 (AR-3).
- The scenario suite tests the engine only through IAssess; the held-out attack generator (iteration 3) is a separate package.
- Relationship ends as in the Student 1 class diagram.

«use»

evaluation (Module 10)

ScenarioSuite

- fixtures: list[ScenarioFixture]

run(engine: IAssess): list[ScenarioResult]
load(path): list[ScenarioFixture]

NFR-Q-01 (7 of 13 at Mid-1), test suite 11

1 — 7..*

«value» ScenarioResult

scenario_id: str
state_ok, rule_ok: bool
lifecycle_ok, explanation_ok: bool
passed: bool {all four}

NFR-Q-01

«value» ScenarioFixture

scenario_id: str
transaction_context
direct_evidence, indirect_evidence
manipulation_risk
expected: ExpectedResult

scenario_fixtures_iteration1.json

1 — 1

«value» ExpectedResult

rule_fired, decision, state
lifecycle_state, hold_reason
explanation facts

NFR-Q-01

«use»

boundary: other modules

«interface» ILifecycleCommands

apply_decision(transaction_id, decision): event

Student 1, Module 3

«use»

«interface» IEventQuery

load_stream(id)
replay(id)
verify_chain(id)

Student 1, Module 3

«use»

«interface» INotify

notify(user, type, payload)

Module 7 stub

«interface» IAssess

assess(request): AssessmentResult

Student 2, Module 5

«use»